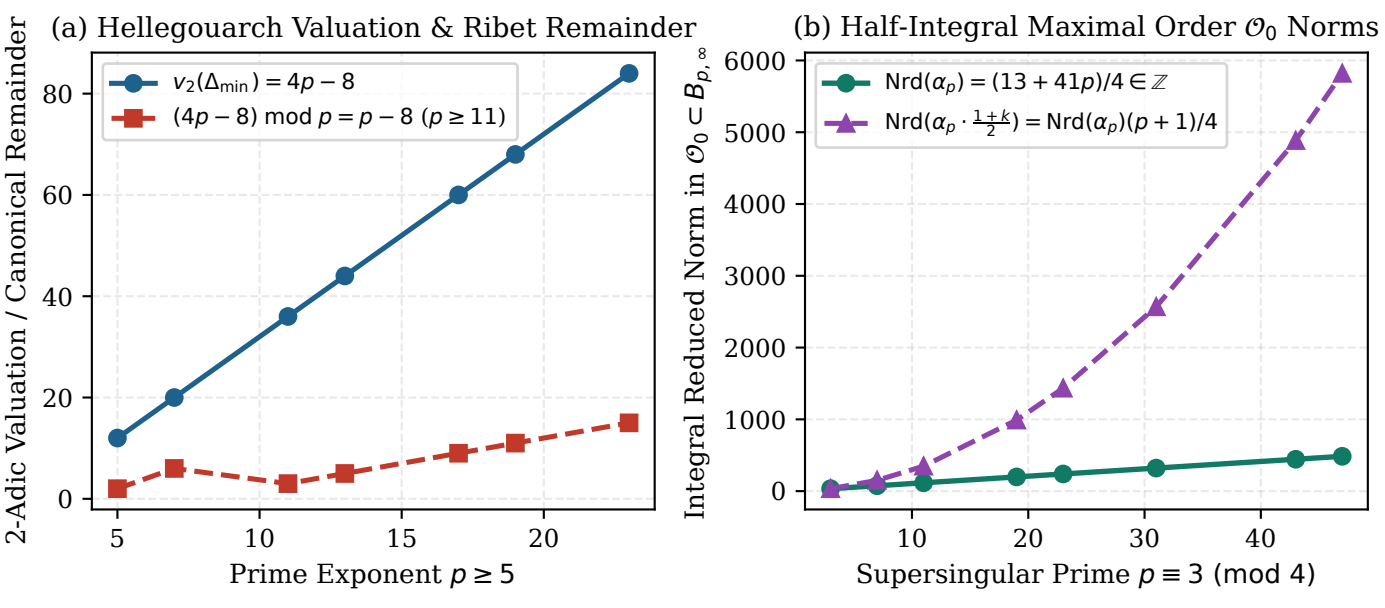
(a) Hellegouarch Valuation & Ribet Remainder
$v_2(\Delta_{\min}) = 4p - 8$
$(4p-8) \bmod p = p - 8$ $(p \geq 11)$
2-Adic Valuation / Canonical Remainder
Prime Exponent $p \geq 5$
0
20
40
60
80
5
10
15
20
(b) Half-Integral Maximal Order $\mathcal{O}_0$ Norms
$\mathrm{Nrd}(\alpha_p) = (13 + 41p)/4 \in \mathbb{Z}$
$\mathrm{Nrd}(\alpha_p \cdot \frac{1+k}{2}) = \mathrm{Nrd}(\alpha_p)(p+1)/4$
Integral Reduced Norm in $\mathcal{O}_0 \subset B_{p,\infty}$
Supersingular Prime $p \equiv 3 \pmod 4$
0
1000
2000
3000
4000
5000
6000
10
20
30
40